

The Abiotic Methane Ceiling Sets a Floor on Detectable Biospheres

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ABSTRACT

Flux based biosignature assessment compares an inferred surface methane flux against a ceiling on abiotic production. The ceiling in current use is a deliberately conservative upper bound, adopted as a single number. It lies more than an order of magnitude above the only other published maximum for the same process, above anything its own generating equation yields at modern Earth crustal production rates, and a factor of three below Earth's own biological methane flux. The choice of ceiling therefore fixes the smallest biosphere an observation can reveal, and the span of defensible choices is wider than the spectroscopic uncertainty of the retrieval it is compared against. That dependence can be reported at no additional computational cost.

1. A BOUND DOING THE WORK OF A LIKELIHOOD

N. F. Wogan et al. (2026) infer surface gas fluxes for an Archean-like TRAPPIST-1e by inverting a coupled photochemical and climate model, then ask whether the inferred fluxes are consistent with life. The step from flux to biosignature runs through a set definition: methane fluxes above 3.7×10^{10} molecules $\text{cm}^{-2} \text{s}^{-1}$ are taken to exceed what ocean bottom serpentinization can supply. That value is the 99.7th percentile of the distribution of maximum abiotic fluxes computed by J. Krissansen-Totton et al. (2018), adopted, as N. F. Wogan et al. (2026) say, for simplicity.

As a set boundary the choice is sound and conservative by design. Its consequences appear at the following step, where the complement of the biosignature region is taken as the abiotic alternative and the resulting odds ratio is used as a proxy for the Bayes factor between life and no life. A bound chosen to be generous then does the work of a likelihood, and its position stops being a matter of caution and becomes a quantitative input.

2. LOCATING THE CEILING

Three comparisons place it, and they are collected in Table 1.

The first does not depend on reading any number as a realised flux. A. Guzmán-Marmolejo et al. (2013) modelled methane production by serpentinization from the kinetics of the reactions involved, with carbon dioxide as the limiting reactant, and report a maximum surface flux of 6.8×10^8 molecules $\text{cm}^{-2} \text{s}^{-1}$ for an Earth mass planet. Two published maxima for the same process differ by 1.74 dex.

The second uses the generating equation itself. The lower corner of the four parameter ranges sampled by J. Krissansen-Totton et al. (2018) is built entirely from Earth calibrated values: their modern crustal production rate, 10 wt% FeO in newly produced crust, the 0.25% conversion of FeO to hydrogen that they calibrate against observation, and the hydrogen to methane ratio of 12 measured in ultramafic hosted vent fluids. Their Equation 6 then returns $0.012 \text{ Tmol yr}^{-1}$, or $\log F_{\text{CH}_4} = 7.67$, a little below the directly measured mid-ocean ridge values. Their distribution runs from what Earth does to the most generous case they were willing to entertain, and the adopted percentile sits 2.90 dex above its lower end.

The third asks what that equation reaches without leaving Earth-like conditions. Holding crustal production at the modern Earth value and setting the other three factors to the generous end of their published ranges, Equation 6 gives $\log F_{\text{CH}_4} = 9.87$. The adopted ceiling is 0.70 dex above that, so it is reachable only by invoking crustal production above modern Earth's. The factor of ten allowed in their range is an early Earth provision, and J. Krissansen-Totton et al. (2018) note that whether such rates are likely for terrestrial exoplanets is an open question. Applied to a planet of roughly Earth size, that provision is carrying much of the ceiling.

Figure 1 sets all of this against published estimates of Earth's realised abiotic methane production (T. L. Schindler & J. F. Kasting 2000; J. F. Kasting 2005; S. Emmanuel & J. J. Ague 2007; J. Fiebig et al. 2007, 2009; M. Cannat et al. 2010; R. S. Keir 2010; D. C. Catling & J. F. Kasting 2017; F. Klein et al. 2019; Y. de Melo Portella et al.

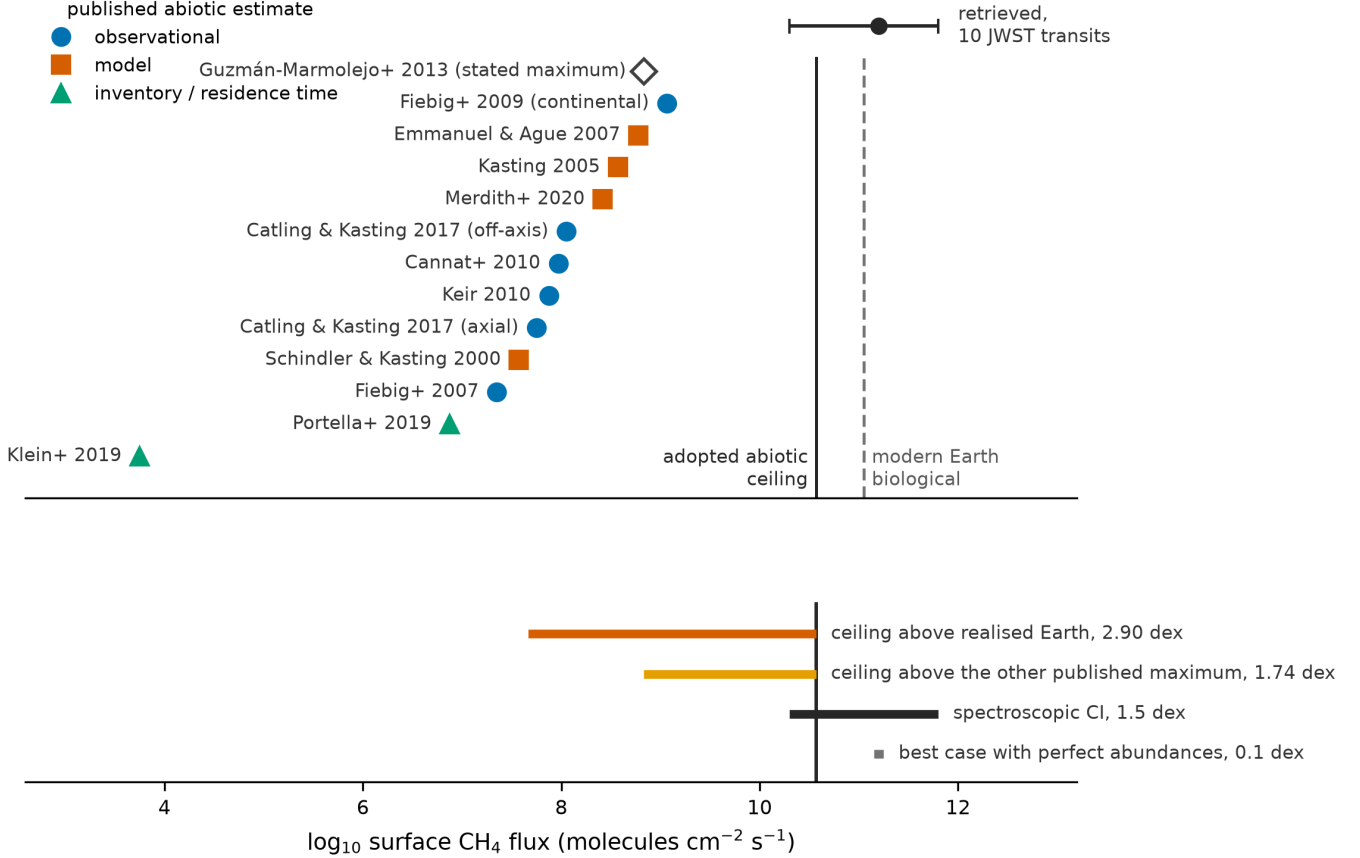


Figure 1. Published estimates of Earth’s abiotic methane production, against the ceiling adopted in flux based biosignature assessment. Filled markers distinguish observational estimates, model estimates, and estimates obtained by dividing a reservoir inventory by a residence time; the open diamond is the maximum reported by [A. Guzmán-Marmolejo et al. \(2013\)](#). The solid line is the adopted abiotic ceiling and the dashed line is modern Earth’s biological methane flux. The point with error bars is the flux retrieved from a synthetic ten transit JWST spectrum of an Archean-like TRAPPIST-1e ([N. F. Wogan et al. 2026](#)). The lower panel compares two framings of the ceiling ambiguity with the credible interval of that retrieval and with the narrowest interval attainable if all gas abundances were known exactly.

2019; [A. S. Merdith et al. 2020](#)), compiled from the supplementary material of [M. A. Thompson et al. \(2022\)](#) and from [A. S. Merdith et al. \(2020\)](#). Every estimate for modern Earth lies below the ceiling, the largest, from continental hydrothermal emissions, by 1.5 dex. Six seafloor estimates, spanning mid-ocean ridges, off-axis vent fields, submarine volcanism and subduction related sites, fall between $\log F_{\text{CH}_4} = 7.3$ and 8.1.

3. WHAT THE CEILING COSTS

A ceiling implies a minimum detectable biosphere. [N. F. Wogan et al. \(2026\)](#) adopt 1.122×10^{11} molecules $\text{cm}^{-2} \text{s}^{-1}$ for modern Earth’s biological methane flux, following [D. C. Catling & J. F. Kasting \(2017\)](#). The ceiling lies a factor of three below that. A planet whose biosphere produced methane at less than a third of Earth’s rate could not be called inhabited on this criterion at any signal to noise. Placing the ceiling at the Earth calibrated value instead moves that floor to roughly one part in 2400 of Earth’s biological flux.

The span is also large next to the measurement it is compared against. The retrieval returns a 68% credible interval 1.5 dex wide for $\log F_{\text{CH}_4}$. Table A1 of [N. F. Wogan et al. \(2026\)](#) gives that width under every combination of gas abundances fixed to their true values, and the narrowest entry is 0.1 dex, the limit of what perfect abundance knowledge would buy on this target. All three framings above, 0.70, 1.74 and 2.90 dex, are comparable to or larger than the interval actually achieved. A posterior width and a range of candidate thresholds are different objects, but both enter the same decision in the same units, and here the geochemical term is the larger.

Table 1. Reference points on the surface methane flux scale.

Quantity	$\log F_{\text{CH}_4}$
Equation 6 at Earth calibrated values (J. Krissansen-Totton et al. 2018)	7.67
Stated maximum, kinetic model (A. Guzmán-Marmolejo et al. 2013)	8.83
Equation 6, modern crustal production, other factors maximal	9.87
Adopted abiotic ceiling (N. F. Wogan et al. 2026)	10.57
Equation 6, all four factors maximal	10.87
Modern Earth biological flux (D. C. Catling & J. F. Kasting 2017)	11.05
Retrieved flux, ten JWST transits (N. F. Wogan et al. 2026)	11.2

NOTE—Fluxes in molecules $\text{cm}^{-2} \text{s}^{-1}$. Every value is recomputed from the cited source rather than read from a figure; see Section 6.

4. THE DEPENDENCE CAN BE REPORTED

None of this is concealed in the original work. N. F. Wogan et al. (2026) state that they adopt a single value for simplicity, note that the inhabited and lifeless flux regions should more realistically overlap, and write down the generalisation in which the sharp set is replaced by a weighting function $p(\text{life} | F)$. What this note adds is the magnitude of the quantity that generalisation would need.

An inexpensive intermediate step is available, and it needs nothing that is not already published. Reading the quoted credible interval endpoints as the 16th and 84th percentiles and 11.2 as the median fixes a split normal, which returns 0.76 for the posterior mass above the adopted ceiling against the 0.77 that N. F. Wogan et al. (2026) tabulate. The same reconstruction puts more than 0.98 of the posterior above the maximum of A. Guzmán-Marmolejo et al. (2013) and above any realised estimate, although below the quoted interval that is an extrapolation and only the qualitative statement is intended. Reporting the biosignature probability as a function of the assumed ceiling, rather than at a single value, therefore costs a line of arithmetic on a posterior that already exists. The corresponding Bayes factor cannot be recovered this way, since it also requires the prior mass implied by the retrieval at each candidate ceiling.

5. LIMITATIONS

The estimates in Figure 1 describe different reservoirs that are only partly additive, and several of the seafloor values are alternative estimates of one quantity rather than separate contributions to it, so the figure should be read as a range of published values and not as a distribution. They also differ in what they measure: the mid-ocean ridge values of R. S. Keir (2010) and M. Cannat et al. (2010) are fluxes released at vents, whereas A. S. Merdith et al. (2020) model production from the iron oxidation state of serpentinite and note that much of it is consumed in the subsurface, which is why their values sit higher. A ceiling for a planet other than Earth would also need its crustal production rate scaled, as A. Guzmán-Marmolejo et al. (2013) do for a five Earth mass planet.

6. SOFTWARE AND DATA

The compiled estimates, the arithmetic reproducing every value quoted above, and the script that draws Figure 1 are available at <https://github.com/ssiddhantsharma/abiotic-ch4>. Values from J. Krissansen-Totton et al. (2018) are recomputed from their Equation 6 and its four published parameter ranges rather than read from their figure, and values from A. Guzmán-Marmolejo et al. (2013) and A. S. Merdith et al. (2020) are taken from those papers rather than from the compilations that cite them.

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